

A Multi-Stream Temporal Context Model for Narrative Memory

Formalizing mechanisms for the equivalence of across-event-bridge and across-event-within-storyline cued recall

1. The puzzle

The pilot data from Xu, Duncan, & Manning (2026) show that cued recall of narratively adjacent but temporally non-adjacent events (across-event-bridge, Experiment 3) is statistically equivalent to cued recall of both temporally and narratively adjacent events (across-event-within-storyline, Experiment 1), with $BF_{01} = 412$.

Under standard TCM (Howard & Kahana, 2002), this result is unexpected. In the grouped condition (Exp 1), consecutive storyline events are encoded back-to-back, so their temporal contexts share maximal overlap. In the interleaved condition (Exp 3), 2–7 intervening events from other storylines separate narratively adjacent events, causing substantial context drift and reducing contextual overlap. TCM would therefore predict *lower* cued recall performance in the bridge condition.

Below we formalize a **Multi-Stream Temporal Context Model (MS-TCM)** that extends TCM with storyline-specific context representations, and show that it naturally predicts the observed equivalence under identifiable parameter regimes.

2. Standard TCM: A brief review

In TCM, a temporal context vector $\mathbf{c}(t) \in \mathbb{R}^d$ is maintained and updated at each encoding step. When item f_i is presented at time t :

$$\mathbf{c}(t) = \rho \cdot \mathbf{c}(t-1) + \beta \cdot \mathbf{c}^{(IN)}_i$$

where:

- $\rho = \sqrt{1 - \beta^2}$ ensures $\|\mathbf{c}(t)\| = 1$ (assuming normalized inputs)
- $\beta \in (0,1)$ controls the drift rate
- $\mathbf{c}^{(IN)}_i$ is the input to context from item i (derived from the item's pre-experimental representation via an association matrix $M^{\{TF\}}$)

During retrieval, a cue item j reinstates a weighted combination of its encoding context and pre-experimental context, which then serves as a probe for retrieving associated items. The

probability of recalling item i given cue j depends on the similarity between the reinstated context and item i 's encoding context.

The temporal contiguity effect arises because items encoded nearby in time share similar contexts. The forward asymmetry arises because the pre-experimental context of item j is only associated (during encoding) with items that follow j , creating an asymmetric retrieval advantage.

3. Multi-Stream Temporal Context Model (MS-TCM)

3.1 Core architecture

MS-TCM maintains *multiple* context vectors during encoding:

1. **Global temporal context** $\mathbf{c}_G(t) \in \mathbb{R}^d$ — drifts at every encoding step, exactly as in standard TCM
2. **Storyline-specific contexts** $\mathbf{c}_S(t) \in \mathbb{R}^d$ for each storyline $S \in \{1, \dots, K\}$ — drifts *only* when an event from storyline S is being encoded

3.2 Encoding dynamics

When event i from storyline S^* is encoded at global time step t :

Global context update (applies at every step):

$$\mathbf{c}_G(t) = \rho_G \cdot \mathbf{c}_G(t-1) + \beta_G \cdot \mathbf{c}_i^{(IN)}$$

where $\rho_G = \sqrt{1 - \beta_G^2}$.

Active storyline context update (only for storyline S^*):

$$\mathbf{c}_{S^*}(t) = \rho_{S^*} \cdot \mathbf{c}_{S^*}(t_{prev}^{S^*}) + \beta_{S^*} \cdot \mathbf{c}_i^{(IN)}$$

where $t_{\{prev\} \wedge \{S^*\}}$ is the most recent time step at which storyline S^* was active, and $\rho_{S^*} = \sqrt{1 - \beta_{S^*}^2}$.

Inactive storyline contexts (for all $S \neq S^*$):

$$\mathbf{c}_S(t) = \mathbf{c}_S(t-1)$$

That is, inactive storyline contexts are "frozen" — they do not drift during encoding of events from other storylines.

3.3 Composite encoding context

Each event i (from storyline S^* , at global time t) is associated with a **composite context**:

$$\mathbf{c}_{comp}(i) = w_G \cdot \mathbf{c}_G(t) + w_S \cdot \mathbf{c}_{S^*}(t)$$

where $w_G + w_S = 1$ are mixing weights that determine the relative contributions of global and storyline-specific context.

3.4 Retrieval dynamics

During cued recall, presentation of cue clip j reinstates a composite context:

$$\mathbf{c}_{ret}(j) = w_G^{ret} \cdot \mathbf{c}_G(t_j) + w_S^{ret} \cdot \mathbf{c}_{S_j}(t_j)$$

Critically, the retrieval weights (w_G^{ret} , w_S^{ret}) can differ from the encoding weights, depending on task demands:

- **Experiment 1 & 2** ("recall what happened immediately before/after in the movie"): Retrieval uses the full composite, so w_G^{ret} and w_S^{ret} are at their default values.
- **Experiment 3** ("recall what happened immediately before/after *within the same storyline*"): The task instruction biases retrieval toward the storyline context, effectively increasing w_S^{ret} relative to w_G^{ret} .

The probability of recalling event i given cue j is:

$$P(i|j) \propto \text{sim}(\mathbf{c}_{ret}(j), \mathbf{c}_{comp}(i)) \cdot a_i$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and a_i captures item-specific factors (e.g., distinctiveness, encoding strength).

4. Deriving the equivalence

4.1 Setup

Consider the two conditions we want to compare:

Across-event-within-storyline (Exp 1, grouped): Events from storyline S are encoded consecutively. Two adjacent events i and $i+1$ from storyline S are separated by zero intervening events from other storylines. Both global and storyline contexts are contiguous.

Across-event-bridge (Exp 3, interleaved): Events from storyline S are separated by m intervening events from other storylines. Global context has drifted substantially, but storyline context has not.

4.2 Context similarity analysis

Grouped condition (Exp 1): For consecutive events $i, i+1$ from storyline S encoded at global times $t, t+1$:

- Global context similarity: $\text{sim}(\mathbf{c}_G(t), \mathbf{c}_G(t+1)) = \rho_G$ (one step of drift)
- Storyline context similarity: $\text{sim}(\mathbf{c}_S(t), \mathbf{c}_S(t+1)) = \rho_S$ (one step of drift)
- Composite similarity $\approx w_G \cdot \rho_G + w_S \cdot \rho_S$ (approximate, for the cross-term analysis)

Interleaved condition (Exp 3): For events $i, i+1$ from storyline S, now separated by m intervening events from other storylines, encoded at global times $t, t+m+1$:

- Global context similarity: $\text{sim}(\mathbf{c}_G(t), \mathbf{c}_G(t+m+1)) \approx \rho_G^{m+1}$ (m+1 steps of drift)
- Storyline context similarity: $\text{sim}(\mathbf{c}_S(t), \mathbf{c}_S(t+m+1)) = \rho_S$ (still only one step of drift within storyline S, since the storyline context was frozen during intervening events)
- Composite similarity $\approx w_G \cdot \rho_G^{m+1} + w_S \cdot \rho_S$

4.3 Condition for equivalence

The two conditions yield equal retrieval probability when:

$$w_G \cdot \rho_G + w_S \cdot \rho_S \approx w_G^{ret} \cdot \rho_G^{m+1} + w_S^{ret} \cdot \rho_S$$

This equality holds when either:

(a) Storyline context dominates ($w_S \gg w_G$): If the storyline context carries most of the weight, then both conditions are dominated by the ρ_S term, which is identical across conditions. The global context term becomes negligible regardless of how much it has drifted.

(b) Task-driven reweighting at retrieval: Even if w_G and w_S are comparable during encoding, the Experiment 3 task instructions ("recall within the same storyline") could increase w_S^{ret} at retrieval. This would amplify the storyline context component and suppress the global context component, again producing equivalence.

(c) Combined with matched interference: In the grouped condition, the event immediately before/after the boundary is from the same storyline but a different scene — introducing within-storyline interference. In the interleaved condition, intervening events are from completely different storylines with different characters, settings, and themes. If this between-storyline material produces *less* interference than within-storyline boundary material (due to greater

distinctiveness), this could partially compensate for the global context drift, narrowing the gap between conditions.

4.4 Numerical illustration

Let $\beta_G = 0.5$, $\beta_S = 0.5$, and $m = 3$ (average number of intervening events in Why Women Kill).

For the grouped condition:

- $\rho_G = \sqrt{1 - 0.25} = 0.866$
- $\rho_S = 0.866$
- Composite similarity = $w_G \cdot 0.866 + w_S \cdot 0.866 = 0.866$

For the interleaved condition:

- $\rho_G^4 = 0.866^4 = 0.563$
- $\rho_S = 0.866$ (only one storyline step)
- Composite similarity = $w_G \cdot 0.563 + w_S \cdot 0.866$

Setting these equal: $w_G \cdot 0.866 = w_G \cdot 0.563$, which requires $w_G = 0$. But we don't need *exact* equality — we need the difference to be undetectable given sample size. Even with $w_G = 0.2$, $w_S = 0.8$:

- Grouped: $0.2 \cdot 0.866 + 0.8 \cdot 0.866 = 0.866$
- Interleaved: $0.2 \cdot 0.563 + 0.8 \cdot 0.866 = 0.806$

The difference (0.060) is modest and may not translate to a detectable difference in hit rates, especially with additional variance from item and participant random effects. With task-driven upweighting of w_S^{ret} in Experiment 3, this gap narrows further.

5. Additional mechanisms

5.1 Context reinstatement at storyline resumption

When a storyline resumes after interruption, the first moments of re-encountering that storyline's characters and setting may trigger **automatic context reinstatement** — reactivating the storyline-specific context to its state from the previous episode of that storyline. This is analogous to the environmental reinstatement effect in episodic memory (Godden & Baddeley, 1975; Smith & Vela, 2001), where returning to the encoding environment improves retrieval.

Formally, when storyline S resumes at global time t after a gap, the storyline context could receive an additional reinstatement boost:

$$\mathbf{c}_S(t) = \rho_S \cdot \mathbf{c}_S(t_{prev}^S) + \beta_S \cdot \mathbf{c}_i^{(IN)} + \gamma \cdot \mathbf{c}_S(t_{prev}^S)$$

where γ captures the reinstatement strength. This would further increase the similarity of storyline contexts across the gap, potentially making them *more* similar than under pure drift.

5.2 Conversational references as explicit reinstatement cues

The Xu et al. (2024) finding that characters refer more to past than future events is directly relevant. When storyline S resumes, characters may reference events from earlier in storyline S — e.g., "Remember when we discussed the project?" — which provides explicit reinstatement of earlier storyline events. This conversational referencing serves as a storyline-specific retrieval cue that bridges the temporal gap.

In MS-TCM, these references can be modeled as additional inputs to the storyline context that bias it toward earlier states:

$$\mathbf{c}_S(t) = \rho_S \cdot \mathbf{c}_S(t_{prev}^S) + \beta_S \cdot \mathbf{c}_i^{(IN)} + \sum_k \alpha_k \cdot \mathbf{c}_{ref_k}^{(IN)}$$

where ref_k indexes events that are explicitly referenced in the current event's dialogue, and α_k controls the reinstatement weight.

5.3 Differential interference from distinct vs. similar intervening material

The release-from-proactive-interference literature (Wickens, 1970) shows that switching to categorically distinct material reduces interference. In the interleaved condition, intervening events from different storylines (different characters, settings, time periods in *Why Women Kill*) are maximally distinct from the target storyline. This may produce less retroactive and proactive interference than within-storyline scene changes (grouped condition), where successive events share characters and thematic content but differ in specific situations.

This can be captured by adding an interference term to the retrieval equation:

$$P(i|j) \propto \text{sim}(\mathbf{c}_{ret}(j), \mathbf{c}_{comp}(i)) \cdot a_i \cdot \exp(-\lambda \cdot I_{ij})$$

where I_{ij} is the cumulative interference from items encoded between i and j , and λ scales the interference effect. If between-storyline events contribute less interference than within-storyline boundary events, then $I_{\text{bridge}} < I_{\text{grouped-boundary}}$, partially offsetting the reduced global context similarity.

6. Testable predictions

The MS-TCM framework generates several predictions beyond the observed equivalence:

6.1 Storyline similarity manipulation

Prediction: The equivalence should break down when storylines are highly similar (same characters, same setting, overlapping themes). With similar storylines, storyline-specific contexts become harder to maintain as distinct representations, reducing the effectiveness of the "frozen context" mechanism.

Test: Use stimuli where storylines share characters or settings (e.g., *The Affair*, which shows the same events from two perspectives) and compare across-event-bridge performance to across-event-within-storyline performance.

6.2 Number of intervening events

Prediction: Under MS-TCM, across-event-bridge performance should be relatively robust to the number of intervening events (m), since storyline context similarity is always ρ_S regardless of m . However, there should be a gradual decline due to the diminishing global context contribution and potential decay of the "frozen" storyline context.

Test: In *This Is Us*, where the number of intervening events varies (1–7, mean 3.8), performance in the bridge condition should show weak or no dependence on m . Standard TCM would predict a strong decline with m .

6.3 Individual differences in working memory

Prediction: Individuals with higher working memory capacity should better maintain and reinstate storyline-specific contexts, thus showing stronger equivalence between bridge and grouped conditions.

Test: Administer a working memory task (e.g., operation span) and test whether working memory capacity moderates the bridge vs. grouped comparison.

6.4 Temporal direction asymmetry across conditions

Prediction: The backward asymmetry observed in the within-event condition ($BF_{10} = 1.6 \times 10^{23}$) but not in the across-event conditions may reflect an interaction between pre-built links and context reinstatement. Within events, pre-built backward links are strong (consistent with Xu et al., 2024) and context overlap is high, so both mechanisms reinforce backward recall. Across event boundaries, pre-built links are weakened (especially across storylines), so the backward advantage from pre-built links is reduced. The storyline context mechanism is approximately symmetric (ρ_S is the same regardless of temporal direction), so it does not introduce directionality.

6.5 Retrieval dynamics: First recall latency

Prediction: Even if hit rates are equivalent between bridge and grouped conditions, first-recall latencies may differ. If retrieval begins with the global context and then incorporates storyline context (a serial or competitive process), bridge condition responses should be slower because the global context is a poorer cue. This latency difference would persist even with equivalent accuracy.

7. Relationship to existing models

7.1 Event Segmentation Theory (EST; Zacks et al., 2007)

EST proposes that ongoing experience is parsed into discrete events, with event models updated at event boundaries. MS-TCM can be seen as a formalization of how event models for different storylines are maintained in parallel, with storyline-specific contexts corresponding to storyline-specific event models.

7.2 Situation models (Zwaan & Radvansky, 1998)

Situation models in narrative comprehension include dimensions like space, time, causation, and character identity. The storyline-specific context in MS-TCM can be interpreted as a compressed representation of the situation model for each storyline. The "freezing" of inactive storyline contexts corresponds to the maintenance of backgrounded situation models.

7.3 Schema-based memory (Baldassano et al., 2018)

Shared narrative schemas within a storyline (e.g., "marital conflict" in *Why Women Kill*) could provide a structural scaffold that makes storyline contexts more robust to interruption. The schema provides a "skeleton" that helps reinstate the storyline context when the storyline resumes.

7.4 Context Maintenance and Retrieval (CMR; Polyn et al., 2009)

CMR extends TCM with source context (e.g., encoding task, list membership). Storyline identity in MS-TCM is analogous to source context in CMR. The key extension is that MS-TCM proposes *independent drift dynamics* for different source contexts, rather than a single context vector with source features.

8. Implementation notes

A full computational implementation would require:

1. **Stimulus representation:** Represent each event as a feature vector (e.g., using sentence embeddings of event descriptions from the video annotations).
2. **Parameter estimation:** Fit $\beta_G, \beta_S, w_G, w_S$ (and optionally γ, λ) to the pilot data using maximum likelihood or Bayesian methods, with the hit rate data from all four conditions simultaneously.
3. **Model comparison:** Compare MS-TCM against (a) standard TCM (single context, β_G only), (b) a "perfect reinstatement" model where storyline contexts are fully reinstated at each storyline switch, and (c) an "independent storylines" model where $w_G = 0$ and storylines are completely independent memory traces.

The critical comparison is between standard TCM and MS-TCM: if MS-TCM provides a substantially better fit (particularly for the bridge vs. grouped equivalence), this supports the multi-stream architecture.

References

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